

OLANDS NORRA UDDE, Sweden

WMO: 025750

Lat: 57.3671N

Lon: 17.0954E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.4	-4.8	-9.9	1.6	-5.0	-8.4	1.8	-3.7	16.3	-1.7	14.9	0.0	5.6	340	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	25.1	19.4	23.5	18.9	22.2	18.2	20.7	23.4	19.8	22.4	18.9	21.5	4.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.4	22.3	18.8	13.6	21.4	17.9	12.8	20.6	59.8	23.5	56.7	22.4	53.8	21.5	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.8	11.2	DB	-7.9	28.1	2.6	1.5	-9.8	29.2	-11.3	30.1	-12.8	30.9	-14.7	32.0	(4)
(5)			WB	-8.5	21.7	2.5	1.5	-10.3	22.8	-11.7	23.6	-13.2	24.5	-15.0	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.8	0.7	2.3	5.6	10.3	15.3	18.5	18.3	14.5	9.4	5.4	2.5
	DBStd	6.98	2.78	3.00	3.02	2.73	3.00	2.64	2.41	2.25	2.34	2.62	2.48	2.96
	HDD10.0	1367	287	261	238	133	33	1	0	0	1	43	138	233
	HDD18.3	3588	545	494	496	381	249	96	27	28	116	277	388	492
	CDD10.0	888	0	0	0	3	42	160	265	258	136	24	1	0
	CDD18.3	67	0	0	0	0	0	5	33	27	1	0	0	0
	CDH23.3	153	0	0	0	0	1	20	89	42	0	0	0	0
	CDH26.7	12	0	0	0	0	0	2	7	3	0	0	0	0
Wind	WSAvg	5.7	6.7	6.3	5.9	5.3	4.8	4.9	4.7	4.8	5.5	6.1	6.4	6.7
Precipitation	PrecAvg	422	29	21	22	25	30	35	43	48	52	36	46	37
	PrecMax	663	67	53	64	72	78	79	102	122	132	119	107	83
	PrecMin	308	1	5	2	5	3	0	4	7	14	3	11	5
	PrecStd	71	17	11	13	15	18	21	25	30	30	24	22	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.3	12.6	15.4	21.3	25.9	27.7	26.5	22.3	16.4	12.1	10.1
		MCWB	6.5	6.3	7.7	10.6	14.2	18.9	21.3	20.7	18.0	13.8	10.7	8.8
	2%	DB	6.6	6.7	9.6	13.1	18.1	23.0	25.6	24.4	20.6	14.8	10.6	8.3
		MCWB	5.4	5.1	6.3	9.2	13.1	17.1	19.8	19.6	17.0	13.1	9.3	7.1
	5%	DB	5.4	5.7	7.5	11.4	16.3	21.1	24.0	23.1	19.2	13.8	9.6	7.2
		MCWB	4.2	4.3	5.2	8.1	12.2	16.3	19.2	19.0	16.0	12.3	8.5	6.2
	10%	DB	4.4	4.6	6.2	9.7	14.8	19.6	22.6	21.9	17.9	12.9	8.7	6.2
		MCWB	3.5	3.3	4.3	7.1	11.4	15.5	18.6	18.4	15.4	11.6	7.8	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.6	8.4	11.4	15.4	19.4	22.5	22.1	19.2	14.5	11.0	8.8
		MCDB	7.6	7.9	11.8	14.6	19.7	23.5	26.0	24.7	21.0	15.6	11.8	9.8
	2%	WB	5.6	5.4	6.7	9.7	13.9	18.0	21.1	20.6	17.6	13.4	9.5	7.3
		MCDB	6.4	6.5	9.0	12.6	17.0	21.7	24.2	23.0	19.8	14.4	10.3	8.0
	5%	WB	4.4	4.4	5.5	8.4	12.8	16.8	20.1	19.8	16.6	12.5	8.7	6.2
		MCDB	5.1	5.5	7.3	10.9	15.6	20.1	22.8	22.2	18.5	13.6	9.3	7.0
	10%	WB	3.6	3.4	4.5	7.2	11.8	15.8	19.2	19.0	15.7	11.7	7.9	5.4
		MCDB	4.3	4.5	6.0	9.3	14.3	18.7	21.8	21.3	17.4	12.7	8.6	6.1
Mean Daily Temperature Range	5% DB	MDBR	2.8	3.2	4.1	4.7	5.5	6.2	5.8	5.5	4.5	3.3	2.6	2.7
		MCDBR	3.6	4.4	6.7	7.9	8.2	8.9	8.1	7.1	6.4	4.5	3.4	3.5
		MCWBR	3.3	3.5	4.4	4.8	4.5	4.7	4.0	3.6	3.6	3.2	3.1	3.3
	5% WB	MCDBR	3.5	4.3	6.4	7.4	7.4	7.9	6.8	6.1	6.0	4.2	3.3	3.3
		MCWBR	3.3	3.6	4.5	4.9	4.5	4.7	3.8	3.5	3.8	3.2	3.0	3.3
Clear-Sky Solar Irradiance	taub	0.299	0.330	0.345	0.382	0.378	0.369	0.398	0.390	0.359	0.349	0.329	0.291	
	taud	2.288	2.325	2.361	2.318	2.407	2.469	2.469	2.406	2.419	2.483	2.412	2.293	2.227
	Ebn at Noon	569	712	808	829	859	872	834	814	790	683	517	457	
	Edh at Noon	49	72	90	109	106	101	106	98	80	66	49	39	
All-Sky Solar Radiation	RadAvg	0.36	0.94	2.46	4.24	5.62	6.14	5.57	4.49	3.04	1.35	0.50	0.26	
	RadStd	0.06	0.20	0.23	0.47	0.43	0.30	0.53	0.39	0.34	0.22	0.09	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.39	N/A	+1.37	N/A	N/A	N/A	-121	-157	N/A	N/A

Nomenclature: See separate page